

1 APPENDIX

1.1 LLM annotations of expanded benchmarks

We prompt an LLM to create the silver labels for the expanded benchmarks (AUTO-FUZZYJOIN +WDC, WEBTABLES +WDC, and FREYJA +WDC), as shown in Figure 3. Given a pair of query column and candidate column, the model is prompted to annotate the joinability as an “equijoin”, “semantic”, or “not joinable”.

1.2 Comparison with ESJ

	MOSAICJOIN	ESJ	MOSAICJOIN _{KMEANS}
Query Column	Top-1 C_M	Top-1 C_E	Top-1 C_K
Pad Abort Test-1 (Apollo)	Pad Abort Test 1		
Soyuz 18a	Soyuz 7K-T No.39		
Venera 1	Venera 3MV-1 No.2		
Spirit (rover)	Spirit rover		
Proba-2	PROBA2		
TDRS-K	TDRS-11	TDRS 3, TDRS 4	
AMC-3		AMC-1 (GE-1), AMC-2 (GE-2), AMC-3 (GE-3)	AMC-4, AMC-9
Galaxy 15		GALAXY 11, GALAXY 12	GALAXY 15, GALAXY 23
Telstar 401		APSTAR 2R (TELSTAR 10), GALAXY 23 (TELSTAR 13)	GALAXY 23 (TELSTAR 13), GALAXY 25 (TELSTAR 5)
	Scores	Scores	Scores
Ranked scores of C_M , C_E , C_K with each method	C_M 0.629 C_E 0.606 C_K 0.603	C_E 0.639 C_M 0.633 C_K 0.599	C_K 0.600 C_M Not in top-50 C_E Not in top-50

Figure 1: Comparison of the rankings of MOSAICJOIN, ESJ, and MOSAICJOIN with KMeans sampling (MOSAICJOIN_{Kmeans}), given a query column about “Artificial Satellites” from the AUTO-FUZZYJOIN +WDC benchmark.

To better understand the performance of MOSAICJOIN compared to ESJ, Figure 1 illustrates an example from AUTO-FUZZYJOIN +WDC where MOSAICJOIN’s k -center sketches produce better rankings than ESJ and MOSAICJOIN with k -means sampling (MOSAICJOIN_{Kmeans}).

Given a query column about artificial satellites, we analyze how each method ranks the same three candidate columns that were each the top-1 column retrieved by each method. MOSAICJOIN’s top-1 column C_M , which is also the expected column in the groundtruth, contains values with a 1-to-1 correspondence with the query column. For example “Pad Abort Test-1 (Apollo)”, “Soyuz 18a”, “TDRS-K” in the query column align with “Pad Abort Test 1”, “Soyuz 7K-T No.39”, “TDRS-11” in C_M , respectively. MOSAICJOIN ranks C_M first (with a score of 0.629) because its k -center sketches maximally covers the column’s embedding space using a fixed-set of diverse representations. Each sketch point captures a distinct region of the column’s embedding space, maximizing the number of query values that have high similarity to sketch vectors. Consequently, MOSAICJOIN ranks C_E and C_K lower because their values, and thus representative points, do not cover as much of the query column’s embedding space as C_M does.

In contrast, ESJ favors larger columns, in this case column C_E (with ~ 380 values) above column C_M (with 72 values). By including all embeddings from candidate columns, ESJ is sensitive to candidate sizes in both Chamfer directions. For directed Chamfer, there is an increased chance that a query value has high similarity with a value from the ~ 380 values in C_E . For inverse Chamfer, many candidate

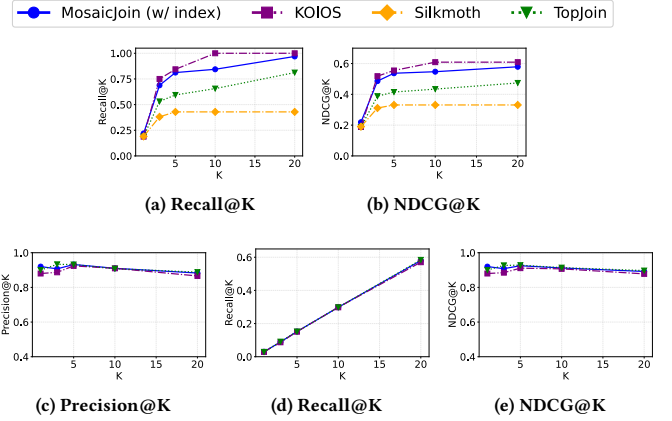


Figure 2: Effectiveness of MOSAICJOIN with an index on WEBTABLES and FREYJA benchmarks, compared to KOIOS, SILKMOTH, TOPJOIN.

values map to the same query value (e.g., “TDRS 3” and “TDRS 4” in C_E both align with “TDRS-K” in the query column), so each such candidate value contributes a high score to the inverse Chamfer score. C_M , with fewer values, has a lower chance for a query value to have a high similarity with a candidate value. Notice that the absolute score of C_E and C_M in ESJ’s rankings are very close, so such a sensitivity to column sizes contributes to the overall ranking.

MOSAICJOIN_{Kmeans} faces a similar challenge. Since its k -means centroids concentrate in dense embedding regions, they miss the distinctive values that make C_M joinable. As a result, C_M does not even appear in MOSAICJOIN_{Kmeans}’s top-50 results. For C_E , the k -means centroids miss many of the ~ 380 values that actually match with query values. Instead, MOSAICJOIN_{Kmeans} ranks C_K first. C_K contains only 55 values, concentrated in a narrow semantic region, of which many values map to one query value (e.g., “AMC-3” in the query column maps to “AMC-4” and “AMC-9” in C_K).

This example illustrates how k -center sketches effectively maximize the coverage of the embedding space. This diverse sampling thus favors columns with 1-to-1 value alignment regardless of data lake column size and number of redundant variations of the same entity, leading to rankings that better reflect value-level joinability.

1.3 Comparison with Index-based approaches

We also compare MOSAICJOIN with SILKMOTH, KOIOS, and TOPJOIN on WEBTABLES and FREYJA benchmarks. WEBTABLES contains only one-to-one value alignments, which are captured by exact bipartite matching methods like KOIOS and SILKMOTH. As shown in Figure 2, MOSAICJOIN with an index outperforms SILKMOTH and achieves similar performance as KOIOS for $k = 20$ on the WEBTABLES benchmark, showing that Chamfer-style scoring is comparable to bipartite matching for one-to-one alignments. FREYJA benchmark contains both one-to-one and one-to-many alignments, and all methods perform comparably (SILKMOTH times out after 10 minutes per query).

Joinability Annotation Prompt

You are an annotation assistant for identifying *joinable attributes*. You will be given rows in CSV format. Each row contains a query attribute and a candidate attribute, along with representative unique values from both columns. Your task is to decide whether the candidate attribute is joinable with the query attribute, and if so, what type of join applies.

Input CSV fields (per row): query_table, query_column, candidate_table, candidate_column, query_unique_values, candidate_unique_values

Task: For each row, compare query_unique_values against candidate_unique_values, using query_table/query_column and candidate_table/candidate_column as context. Decide whether the two attributes are joinable.

Output format: Return a CSV with *exactly* this header:

query_table,candidate_table,joinability,confidence,analysis

Joinability labels: joinability must be one of {semantic, equijoin, not joinable}.

- **equijoin:** Values represent the same identifier/value space and can match directly (allowing minor normalization such as case, whitespace, or punctuation).
- **semantic:** Values are not usually identical but there is a stable mapping/transformation enabling a meaningful join (e.g., city→state, county→state, code→name).
- **not joinable:** No stable direct match or meaningful semantic mapping is supported by the values.

Confidence: confidence is a real number in $[\emptyset, 1]$.

Analysis: analysis must be *one short, precise sentence* stating the key evidence (e.g., strong overlap, same entity type, hierarchical mapping, incompatible domains).

Rules:

- **Use values as primary evidence:** Prefer what the values represent; use column/table names only as supporting context.
 - **Be conservative:** If evidence is weak or ambiguous, choose not joinable.
 - **One output row per input row:** Preserve the input order exactly.
 - **Output only CSV text:** No markdown, no extra commentary.
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Figure 3: Joinability annotation prompt used to label candidate attribute pairs as equijoin, semantic, or not joinable.